



RKDF UNIVERSITY, BHOPAL

B.Sc.(Physics)

SCHEME

Semester – I to VI

No	Subject Code	Subject Title	Marks Allotted						
			Assignment Marks		Theory Marks		Practical Marks		Total Marks
			Max	Min	Max	Min	Max	Min	
1	BPY 111	MECHANICS	20	8	80	27	50	17	150
2	BPY 121	PROPERTIES OF MATTER AND SOUND	20	8	80	27	50	17	150
3	BPY 131	ELECTRICITY AND MAGNETISM	20	8	80	27	50	17	150
4	BPY 141	ELECTRICITY AND MAGNETISMBASIC ELECTRONICS	20	8	80	27	50	17	150
5	BPY 151	NUCLEAR PHYSICS	20	8	80	27	50	17	150
6	BPY 161	ELECTRONIC COMMUNICATION	20	8	80	27	50	17	150



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Syllabus Semester – I

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc (Physics)	MECHANICS	BPY 111

UNIT I

Projectile: Definition of Range, time of flight and angle of projection - Range up and down an inclined plane maximum range - two directions of projections for a given velocity and range. Impulse-Impact: Laws of impact – coefficient of restitution – impact of a smooth sphere on a fixed smooth plane – Direct impact between two smooth spheres – Loss of kinetic energy in direct impact – velocity change in oblique impact between two smooth spheres.

UNIT II

SHM: Composition of two SHM's of same period along a straight line and at the right angles to each other Lissajous figures. Dynamics of Rigid Bodies: Compound pendulum theory condition – for minimum period interchangeability of center of suspension and center of oscillation – g using compound pendulum - Bifilar pendulum – parallel and non – parallel threads.

UNIT III

Center of gravity: Center of gravity of a solid cone, Solid hemisphere, hollow hemisphere and a tetrahedron. Friction: Laws of friction - angle of friction – resultant reaction and cone of friction – equilibrium of a body on an inclined plane under the action of a force.

UNIT IV

Center of pressure: Definition – center of pressure of a rectangular lamina and triangular lamina. Hydrodynamics: equation of continuity of flow – Bernoulli's theorem – venturimeter – Pitot's tube. Conduction and Radiation : Thermal Conductivity - definition - thermal conductivity of a bad conductor - Lee's disc method - good conductor - Searle's method - radiation - Blackbody radiation – definition - Wien's Displacement law - Rayleigh Jean's law - Planck's law - Stefan's law and experimental verification of Stefan's law - Solar constant - temperature of the sun - by Angstrom's Pyrheliometer.

UNIT V

Maxwell's Thermodynamic relations : Derivation of Maxwell's thermodynamic relations - Helmholtz function - Gibbs function - Enthalpy – T-ds equation - Clausius - Clapeyron's Latent heat equation - specific heat relations.

Books for Study:

1. Mechanics and Mathematical Methods by R. Murugesan. S.Chand and Co.
2. Dynamics by M. Narayanamurthi and M. Nagarathnam, The National Publishing Company.
3. Statics, Hydrostatics and Hydrodynamics By Narayanamurthi and M. Nagarathnam, The National Publishing Company.
1. Classical Mechanics by H. Goldstein Addison Wesley Publications 2. Mechanics by D.S. Mathur, S. Chand and Co.,



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SEMESTER-I

PRACTICALS

1. Young's Modulus (q) – Non uniform Bending – pin and microscope method. Determination of unknown mass of an object.
2. Young's Modulus (q) – Non uniform bending – scale and telescope method. Determination of unknown mass of an object.
3. Torsion pendulum – Rigidity Modulus.
4. Surface tension and interfacial surface tension – Drop Weight method.
5. Compound pendulum – Determination of g and k .
6. Sonometer – frequency of a tuning fork – Determination of mass of a stone.
7. Sonometer – R.D of a solid and liquid.
8. Spectrometer – (i-d curve).
9. Spectrometer - Grating – normal incidence measurement of Wavelength.
10. Potentiometer – calibration of low range Voltmeter.
11. P.O. Box – Temperature coefficient of resistance.
12. Lee's Disc – Thermal conductivity of a bad conductor and emissivity.
13. Joule's calorimeter – Specific heat capacity of a liquid – Barton's correction.

Reference books :

1. Practical Physics – Ouseph, Srinivasan & Vijayendran.
2. Practical Physics – P. R. Sasi Kumar, PHI.
3. Advanced Practical Physics – S. P. Singh, Pragathi Prakasam.
4. Practical Physics – St. Joseph College, Trichy.



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Syllabus Semester – II

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc (Physics)	PROPERTIES OF MATTER AND SOUND	BPY 121

UNIT-I

Elasticity: Three types of elastic moduli - Poisson's ratio - Bending of beams - Expression for bending moment - Depression of the loaded end of a Cantilever - uniform - non uniform bending - theory - experiment pin and microscope method - work done in uniform bending – Koenig's method – non-uniform bending - theory - expression for couple per unit twist - determination of rigidity modulus - Static torsion method with scale and telescope - Rigidity modulus by torsion pendulum with mass.

UNIT-II

Interference and Interferometers : Coherence - temporal coherence and spatial coherence - Air wedge – testing the planeness of a surface - Michelson Interferometer - types of fringes - Difference in wavelength of Sodium D1, D2 lines and thickness of a thin transparent plate. Multiple beam interference - Feby - Perot interferometer - formation of fringes. Holography : Holography - recording and reconstruction.

UNIT-III

Surface tension and Osmosis : Surface energy - angle of contact and its determination - excess of pressure inside curved surface - formation of drops - Experimental study of variation of Surface tension with temperature - drop weight method of determining surface tension and interfacial surface tension - angle of contact of mercury - Quincke's method - surface tension and vapour pressure osmosis - experimental determination of osmotic pressure - Laws of osmosis pressure - osmotic and vapour pressure of a solution.

UNIT-IV

Sound: Definition of free, damped and forced vibrations – Theory of forced vibrations - Resonance - Sharpness of resonance - Fourier's theorem - application for Saw- tooth wave and square wave. - Sonometer - determination of A.C. frequency using sonometer - Determination of frequency using Melde's apparatus.

UNIT-V

Ultrasonics: Ultrasonics - Production - Piezo electric method – magneto-striction method - detection - properties - applications. Acoustics : Acoustics of buildings - reverberation time - derivation of Sabine's formula - determination of absorption coefficient.

BOOKS FOR STUDY:

1. Elements of properties of matter by D. S. Mathur S. Chand & Co., (2005).
2. Properties of matter by R. Murugesan, S. Chand & Co., (2005).
3. Properties of matter by Brijlal and N. Subramaniam S. Chand & Co., (2005).
4. Properties of matter and Acoustics by R. Murugesan, S. Chand & Co., (2005).
5. A Text Book of Sound by N. Subramaniam and Brijlal, S. Chand & Co., (2005).



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BOOKS FOR REFERENCE:

1. Fundamentals of General Properties of Matter, H. R. Gulati, S. Chand & Co., (2005).
2. Properties of Matter, Subramania Iyer and Ranga Rajan, Viswanathan Publication (2002).
3. A Text Book of Sound (2005), R. L. Saighal, S. Chand & Co.,
3. Geometrical and Physical Optics - P. K. Chakrabarti, New Central Book Agency (P) Ltd, Kolkata., 2005.
4. Optics - D.R. Khanna and H.R. Gulati, R. Chand & Co, New Delhi., 1979.
5. Engineering Physics - G. Vijayakumari, Vikas Publications.

- : 1. Optics - Eugene Hecht, Fourth Edition, Pearson Education, New Delhi. 2007.
2. Fundamentals of Optics - Jerkins A Francis and White E Harvey, McGraw Hill Inc., New Delhi, 1976.
 3. Optical Physics - S.G. Lipson, H. Lipson and D.S. Tannhauser, Cambridge University Press. 1995.
 4. Fundamentals of Optics - M.G. Raj, Anmol Publications Pvt. Ltd., New Delhi, 1996.

SEMESTER-II

PRACTICALS PRACTICAL

01. Young's modulus (q) - uniform bending - pin and microscope.
02. Young's modulus (q) - uniform bending - scale and telescope method.
03. Static Torsion – Rigidity modulus.
04. Torsion Pendulum – Moment of Inertia and Rigidity modulus – symmetrical masses.
05. Coefficient of Viscosity of a liquid - graduated burette - radius by mercury pellet method.
06. Melde's apparatus - frequency - transverse and longitudinal modes.
07. Specific heat capacity of a liquid by cooling - verification of Newton's law of cooling.
08. Air Wedge - thickness of a wire and its insulation.
09. Spectrometer - grating - minimum deviation – Determination of wavelength of mercury lamp.
10. Potentiometer - ammeter calibration.
11. Potentiometer - Specific resistance of the given coil and length of second coil without unwinding.
12. M and BH - Deflection Magnetometer - $TAN A$ and $TAN B$ position.
13. Field along the axis of a coil - deflection magnetometer – determination of BH .
14. Carey-Foster's bridge - Specific resistance of a coil.
15. BG - Comparison of Capacities.
16. BG - Comparison of EMF's of two cells.
17. Zener diode – Voltage regulator using four diodes and percentage of regulation.
18. Verification of De Morgan's theorem.



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Syllabus

Semester – III

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc (Physics)	ELECTRICITY AND MAGNETISM	BPY 131

UNIT- I

Principle of a capacitor - energy stored in a capacitor - energy density - change in energy due to dielectric slab - force of attraction between plates of a charged capacitor - capacitance of a spherical and cylindrical capacitors - types of capacitors - electrometers - Kelvin's attracted disc electrometer - quadrant electrometer - measurement of potential, ionization current and dielectric constant (solid).

UNIT - II

Positive ray analysis - Thomson's parabola method - theory - determination of e/m and mass of positive ions - Aston's mass spectrograph determination of masses of isotopes - uses of mass spectrographs – separation of isotopes - mass spectrograph method - diffusion method - thermal diffusion method - pressure diffusion method.

UNIT - III

Theory of alpha scattering - Rutherford scattering formula – experimental verification - nature of privileged quantum orbits - Bohr's correspondence principle - effect of motion of nucleus - evidences in favour of Bohr's theory - Determination of critical potential - Davis and Goucher's method - Sommerfield's relativistic atom model -application to fine structure of $H\alpha$ line.

UNIT -IV

Transient current - growth and decay of current in a circuit containing resistance and inductance - growth and decay of charge in a circuit containing resistance and capacitance - measurement of high resistance by leakage - growth and decay of charge in a LCR circuit - condition for the discharge to be oscillatory - frequency of oscillation - Importance in wireless telegraphy.

UNIT- V

Alternating current - peak, average and RMS value of current and voltage - form factor - j operator - ac circuit containing resistance and inductance - choke coil - ac circuit containing resistance and capacitance - series and parallel resonance circuits - Q factor - power in an ac circuit containing LCR – Wattless current - Transformer - construction, theory and uses - energy loss - skin effect – Tesla coil.

Books for Study:

1. Electricity and Magnetism - Brijlal and Subramaniam, S. Chand & Co., (2005).
2. Electricity and Magnetism - R. Murugesan, S. Chand & Co., (2005).

Books for Reference:

1. Electricity and Magnetism - D. N. Vasudeva, S. Chand & Co., (2005).
2. Electricity and Magnetism- K. K. Tewari, S. Chand & Co., (2005)



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Syllabus

Semester – IV

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc (Physics)	ELECTRICITY AND MAGNETISMBASIC ELECTRONICS	BPY 141

UNIT I

Special diodes : Light Emitting Diode (LED) and its advantages – multicolor LEDs and its applications - Photo diode – characteristics and applications – Tunnel diode and its characteristics – Tunnel diode as an Oscillator – Varactor diode – Theory and its applications – Shockley diode – PIN diode and its applications.

UNIT II

Hybrid (h) parameters - determination of h -parameters – h -parameters equivalent circuit – performance of a linear circuit in h -parameter – the h -parameter of a transistor – Nomenclature for transistor h - parameters – input impedance, voltage gain and current gain in h - parameters – experimental determination of h -parameters – limitations of h -parameters.

UNIT III

Common emitter transistor as an amplifier - DC and AC load line analysis - Transistor biasing - stabilization - base resistor method - feed back resistor method - Voltage divider bias method - Construction of JFET – its characteristics and parameters - Common source JFET amplifier - MOSFET - Depletion MOSFET- Enhancement MOSFET - UJT, SCR - Construction, working, V-I characteristics and their application.

UNIT IV

Multistage amplifier – definition of gain, frequency response, decibal gain and bandwidth – operation, frequency response, advantage, disadvantage and applications of RC coupled CE transistor amplifier (two stage) and transformer coupled amplifier.
Principle of feedback in amplifiers – positive and negative feedback – effect of negative feedback - emitter follower – positive feedback amplifier as an oscillator - Hartley oscillator, Wien-bridge oscillator and Piezo electric crystal oscillator.

UNIT V

Multivibrators - astable, monostable and bistable multivibrator using transistor. Operational Amplifier : Differential amplifier - basic circuit and its operation –CMRR – Op-amp – Block diagram and explanation – applications – differentiator , integrator and comparator – multistage op-amp – solving simultaneous equations.

Books for Study:

1. Basic Electronics (Solid state), B.L. Theraja, S. Chand & Co., (2000).
2. Principles of Electronics, Metha, V.K. S. Chand & Co., (2001).
3. Elements of electronics, M.K. Bagde and S.P. Singh, S. Chand & Co. (1982).
4. Foundations of electronics, D. Chattopadhyay and etal., New Age International Publishers (1999).
5. Hand book of Electronics - Gupta & Kumar, Pragati Prakhasan (2005).



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Syllabus Semester – V

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc (Physics)	NUCLEAR PHYSICS	BPY 151

UNIT I

RADIOACTIVITY:

Laws of successive disintegration - transient - and secular equilibria - range of alpha particles - experimental measurement - Geiger - Nuttall Law - alpha ray spectra - Gamow's theory of alpha decay and its experimental verification - Beta ray spectra - origin of line and continuous spectrum - Fermi's theory of beta decay - K electron capture - Nuclear Isomerism.

UNIT II

NUCLEAR DETECTORS:

Principle and working - solid state detector - proportional counter - Wilson's cloud chamber - Scintillation counter.

Accelerators : Synchrocyclotron - Synchrotron - Electron synchrotron - proton synchrotron - Betatron.

UNIT III

ARTIFICIAL TRANSMUTATION:

Rutherford's experiment - Bohr's theory of Nuclear disintegration – Q value equation for a nuclear reaction - threshold energy - types of nuclear reaction - energy balance and the Q value - threshold energy of an endoergic reaction.

Neutron: Mass, charge, decay, spin and magnetic moment, Neutron diffraction, absorption of neutron by matter - neutron sources - detectors - neutron collimator.

UNIT IV

NUCLEAR STRUCTURE: General properties of nucleus – size , mass and charge.

Proton - electron theory - proton - neutron theory - Nuclear size - experimental measurement of nuclear radius - mirror nuclei method - meson theory of nuclear forces - nuclear models - liquid drop model - Weizacker's semi - empirical formula - nuclear shell model.

UNIT V

NUCLEAR FISSION, FUSION AND ELEMENTARY PARTICLES

Nuclear fission - Bohr Wheeler theory - chain reaction – critical size and critical mass - Nuclear fission reactor – Nuclear fusion - source of stellar energy - Carbon – Nitrogen cycle - Proton – Proton cycle - Thermo Nuclear reaction – plasma.

Elementary Particles - types of interactions - classification of elementary particles - particle quantum numbers – baryon number - lepton number - strangeness number - hyper charge -sospin quantum number.

Books for Study:

1. Modern Physics by R. Murugesan, S.Chand & Co., (2005)
2. Atomic Physics by J.B. Rajam, S.Chand & Co., (2005)
3. Nuclear Physics by D.C.Tayal, Himalaya Publishing House.



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Books for Reference:

1. A Source book on Atomic energy by Samuel Glass Stone (2002)
2. Atomic and Nuclear Physics by Albright Semat (2003)
3. Atomic and Nuclear Physics by Little field and Thorley. ELBS (2002)
4. Basic Nuclear Physics and Cosmic rays, B.N. Srivatsava, Pragti Prakasham.



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Semester – VI

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc (Physics)	ELECTRONIC COMMUNICATION	BPY 161

UNIT I

Modulation - definition - types of modulation AM, FM, PM – expression for amplitude modulated voltage - wave form of amplitude modulated wave - collector modulation circuit - single side band generation - balanced modulator - AM transmitter - block diagram and explanation - frequency modulation - expression for frequency modulated voltage - side bands in FM, AM production by transistor modulator - comparison of AM, FM, PM.

UNIT II

Demodulation - definition - diode detection of AM signals - FM detection - Foster Seely discriminator - radio receivers - straight receivers - TRF receivers - super heterodyne receivers - block diagram - explanation of each stage - FM receivers – block diagram.

UNIT III

TV - plumbicon - vidicon - scanning and interlaced scanning – block diagram of TV transmitter and receiver - colour TV - generation R, G, B signals - simplified block diagram of colour TV transmitter and receiver – TV transmitting antennas - dipole panel - TV receiving antenna - Yagi antenna - log antenna - log periodic antenna.

UNIT IV

RADAR - principle of radar - Radar equation - radar – transmitting systems - radar antennas - duplexer - radar receivers uses of radar – opto-electronic devices – photoconductive cell - solar cell - phototransistor – LED and LCD - construction and working.

UNIT V

Digital communications - digital technology - fundamentals of data communication systems - characteristics of data transmission circuits – digital codes - error detection and correction - data sets and inter connection - requirements - modern classification - modern interfacing.

Books for Study:

1. Hand book of Electronics - Gupta & Kumar, Pragati Prakhasan (2005).
2. Electronics Communication Systems - Kennedy and Davis, TMH.
3. Electronics Communications – Roody Coolen, Pearson Education Publication.
4. Electronics Communications - Frenzel, TMH.